

AI-Powered DevOps: Unlocking Delivery Excellence

Leveraging AI To Accelerate Product Growth, Optimize Workflows,
And Reduce Overall Technical Debt

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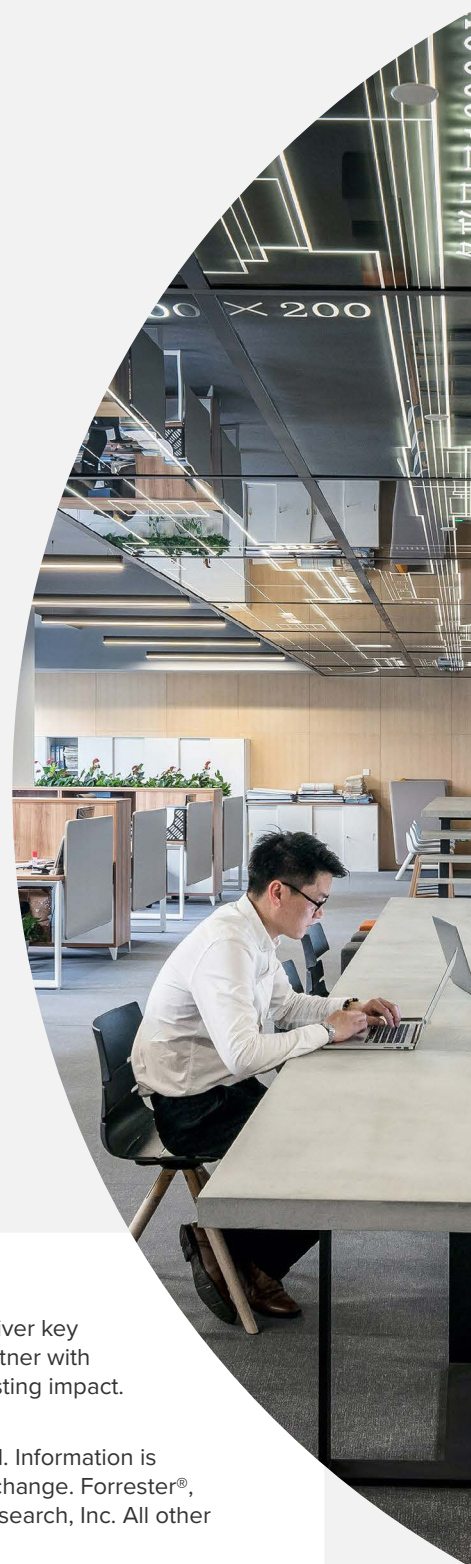
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Executive Summary

DevOps environments are complex, with scattered systems, documentation, and knowledge making them ideal for large language models to streamline. Traditional workflows waste time and resources by diverting skilled developers from high-value tasks to repetitive ones like testing, configuration, and incident triage.

To address these challenges, organizations are prioritizing strategies to improve operational efficiency and resilience by shifting their approach to DevOps and application modernization with AI-driven solutions. However, in today's fast-moving landscape, adopting these tools is no longer optional — it's essential to stay competitive, meet evolving market demands, and avoid falling behind. They must strike a balance between innovation, robust security, streamlined workflows, and cost management before competitors seize the advantage.

Achieving this balance, however, is easier said than done. Our study found that although AI and DevOps are recognized as powerful drivers of productivity and innovation, more than half of respondents stated their DevOps team spends at least a full workday identifying where code changes need to be made. AI integration is also a challenge due to high costs and complex processes.

In December 2025, AWS Marketplace commissioned Forrester Consulting to evaluate how organizations are implementing AI in the software development lifecycle (SDLC) and DevOps practices and identifying barriers preventing their full adoption. Forrester conducted an online survey with 320 respondents who are responsible for their organization's DevOps and AI decision-making to explore this topic.

